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- **TY BTECH CE**
- **Optimization Techniques Experiment – 07 (Multi-objective Optimization)**

Aim

1. Implement all 37 multiobjective optimization algorithms from scratch and explain the working of each algorithm step by step.

2. Describe the advantages and disadvantages of each algorithm and support the analysis with experimental plots and numerical results.

3. Select suitable algorithms for given problems by considering convexity, number of objectives, variable types, constraints, preferences, and evaluation budget.

4. Evaluate solution-front quality using Hypervolume (HV), Inverted Generational Distance (IGD), and Spacing.

Google Drive Folder:

https://drive.google.com/drive/u/5/folders/1e3FpToEx577iFqscnc4vNQyxMnbzD3NN

Task C Algorithms (Roll No. 22)

Algorithm No.	Algorithm
22	MOPSO - Multi-Objective Particle Swarm Optimization
34	HypE - Hypervolume Estimation Algorithm
9	VIKOR - Multi-Criteria Compromise Ranking Method